

Phase-space validation

Engineering Phase-space / dialectic

NeuralCompose

2026-09-23

Document Control

Version	1.0
Prepared By	Automated local engineering validation
Reviewers	Not assigned
Date Prepared	2026-09-23
Reporting Window	Fixed benchmark schedule; see run manifest
Status	Measured engineering report

Purpose

Measure whether the full composition fits the frame budget, against trace-only rendering on identical deterministic input. Frame timing is limited by display refresh, so it cannot resolve rendering cost below one refresh interval.

Keep per-run evidence, build identity, and limits available to maintainers.

Engineering Context

The scene combines four-channel EEG, frequency contributions, model candidates, and final text.

Semantic position encodes chronology and role only. Explicit edges carry cosine; absent embeddings do not produce invented coordinates or edges.

Experiment Summary

Field	Details
Experiment Name	Phase-space / live dialectic renderer
Experiment Key	phase-space-v1

Owner Team	NeuralCompose
Business Owner	Not specified
Product Surface	Linux native Rust GPU window
Primary Objective	Measure scene-submission timing and preserve signal/model semantics
Control (Variant A)	Trace-only rendering
Treatment (Variant B)	Trace, band atmosphere, chronological semantic graph, final text
Allocation	Three rounds; every case once per round
Unit of Randomization	Recorded seeded order within round
Audience	Engineering maintainers; no human participants
Exclusions	Dirty, incomplete or aborted runs are non-quotable; no replacements
Start Date	2026-09-23T08:03:13.485444+00:00
End Date	2026-09-23T08:42:29.413207+00:00
Planned Runtime	36 × (5 s warmup + 60 s): 39 minutes, plus setup
Actual Runtime	39.27 minutes, including warmup and process setup

Design Notes

Three complete rounds cover twelve cases. Each fresh process warms for five seconds and measures for sixty. Order is shuffled with a recorded fixed seed.

Both variants receive identical synthetic EEG and semantic events. This tests engineering behavior, not human perception or biological efficacy.

Hypothesis

The full composition can satisfy the registered submission-cadence and receive-to-submit targets while preserving the core and dialectic contracts.

Success Criteria

Apply both timing targets separately to every run. Only three successful quotable repeats accept a case; mixed results remain unresolved.

Metric	Target/Rule
Primary Metric	Per-run nearest-rank frame-interval p95 ≤ 20 ms
Guardrail 1	Per-run sample-to-submit age p95 ≤ 100 ms
Guardrail 2	Bounded buffers; no rendering-induced ingestion loss
Guardrail 3	Observer parity under a full publisher channel; absence never becomes similarity zero
Decision Rule	Three quotable repeats: 3/3 accepted; 0/3 not met; mixed or missing unresolved. No extra runs.

Pre-Launch Validation

- Require a clean committed checkout and release executable from that checkout.
- Freeze the schedule, fixtures, variants, and decision rule before recorded execution.
- Capture the executable digest, GPU/driver, actual scene extent, display configuration, and per-run timing arrays.
- Keep dirty or incomplete runs non-quotable; never replace them.
- Inspect preflight images and run deterministic tests, including forced publisher backpressure, before freezing the build.

Sample and Exposure Summary

36/36 runs available. Cases: 12 accepted, 0 targets not met, 0 unresolved.

Metric	Variant A (Control)	Variant B (Treatment)
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Completed runs	18	18
Rendered frames	129111	129059
Accepted samples	276424	276440
Cases × repeats	6 × 3 planned; 18 runs	6 × 3 planned; 18 runs
Sample Ratio Check	Not applicable; fixed engineering schedule	Not applicable; fixed engineering schedule

Primary Outcome

Measure	Variant A	Variant B	Absolute Delta	Relative Delta
Worst run-level frame p95 (ms)	10.43	10.03	-0.40	Not a pooled percentile
95% Confidence Interval	Not applicable	Not applicable	Descriptive repeats	Not applicable
Two-Sided p-value	Not applicable	Not applicable	No inferential test	Not applicable
Outcome Narrative	Trace-only	Full composition	36/36 runs available. Cases: 12 accepted, 0 targets not met, 0 unresolved.	This host/build only

The primary table reports the worst run-level p95 in each variant, not a pooled p95. Case spreads below retain the three-repeat variation.

Guardrail Metrics

Guardrail Metric	Variant A	Variant B	Delta	Status
Worst sample-to-submit p95 ms	40.97	40.82	-0.16	Per-run target ≤100 ms
Maximum clipped display fraction	0.05	0.05	Descriptive	Impulse fixture intentionally exceeds scale

Observer / drift parity	Deterministic test suite	Full-channel barrier test	No model/log change permitted	See validation transcript
Signal gap / missing similarity	Deterministic tests	Deterministic tests	No invented continuity or edges	See validation transcript

Timing thresholds apply per run. Other rows identify deterministic invariants and descriptive signal properties; they do not imply physiological validation.

Segment Results

The six fixture/mapping groups were specified in advance. Each table entry is min / median / max of three run-level frame p95 values, in milliseconds.

Segment	Control Value	Treatment Value	Absolute Delta	Interpretation
rhythmic/channels	9.30 / 9.36 / 10.09	9.00 / 9.67 / 10.03	0.32	A: accepted; B: accepted
rhythmic/delay	9.09 / 9.25 / 9.33	9.07 / 9.31 / 9.63	0.06	A: accepted; B: accepted
noise/channels	9.43 / 9.49 / 9.86	9.14 / 9.17 / 9.90	-0.32	A: accepted; B: accepted
noise/delay	9.21 / 9.66 / 10.43	9.09 / 9.35 / 9.38	-0.32	A: accepted; B: accepted
impulse/channels	9.36 / 9.44 / 9.51	9.11 / 9.17 / 9.61	-0.27	A: accepted; B: accepted
impulse/delay	9.14 / 9.16 / 9.81	9.39 / 9.56 / 9.65	0.40	A: accepted; B: accepted

Report every case even when targets fail. No extra runs resolve a mixed outcome.

Data Quality and Limitations

- Desktop scheduling and thermal/load conditions can affect frame timing. The recorded host configuration bounds the claim.

- No human allocation, engagement, retention, or cognitive-state outcome was measured.
- Capture incomplete runs in the failures file; keep them unresolved and unreplaced.
- Only Linux on the named GPU/driver is measured. No other support-matrix row is promoted.
- Sample-to-submit age begins at viewer receipt and ends after scene queue submission; it is not sensor-to-screen latency.

Historical text logs have no embeddings. They cannot reconstruct spatial similarity or a missing model trajectory.

Interpretation

36/36 runs available. Cases: 12 accepted, 0 targets not met, 0 unresolved.

The result supports only the reported engineering decision for this host/build. It does not establish a universal performance guarantee or a cognitive intervention.

Decision Log

Item	Decision
Final Decision	36/36 runs available. Cases: 12 accepted, 0 targets not met, 0 unresolved.
Decision Date	2026-09-23
Approvers	Not assigned; this report records measurements, not approval
Rollout Type	Separate opt-in visualization shell; no model-policy change
Rollback Trigger	Any observer parity, source freshness, or provenance violation
Follow Up	Unresolved cases remain unresolved. Any follow-up requires a new protocol.

Post Test Actions

1. Retain all run files and the registered schedule.
2. Use the dashboard to inspect each case and all three repetitions.
3. Keep mixed outcomes unresolved. Specify any follow-up before collecting it.
4. Preserve observer parity when extending the live feed.
5. Archive the build digest, source revision, template hashes, rendered previews, and validation transcript together.

Appendix A: Metric Definitions

- **Frame interval:** monotonic milliseconds between phase-scene GPU queue submissions; nearest-rank p95 per run.
- **Sample-to-submit age:** submission time minus receive time of the newest accepted EEG frame.
- **FPS:** observed scene-submission intervals divided by their elapsed measurement span.
- **Live fraction:** submissions marked Live divided by submissions in the measurement span. Clipping fraction counts out-of-range vertices across rendered windows.
- **Case verdict:** three quotable runs required; 3/3 meeting both targets accepted, 0/3 not met, mixed or incomplete unresolved.

Appendix B: Analyst Notes

- Use descriptive min/median/max of run-level p95 values; no pooled p95 or inferential significance test.
- The registered count is fixed at three runs per case; no optional stopping, replacement, or extension runs.

- Build revision: 3745369692fdd9908588316bac212f6e3b171183. Binary identity and hardware are in Runs and the benchmark manifest.
- Original source templates are unchanged; hashes accompany the artifacts. Missing measurements and unassigned reviewers remain explicit.

Per-case sample-to-submit p95 spread (ms; min / median / max)

rhythmic-channels-trace: 40.28 / 40.54 / 40.71

rhythmic-channels-full: 40.60 / 40.61 / 40.64

rhythmic-delay-trace: 40.27 / 40.44 / 40.46

rhythmic-delay-full: 40.48 / 40.58 / 40.67

noise-channels-trace: 40.55 / 40.58 / 40.61

noise-channels-full: 40.51 / 40.51 / 40.82

noise-delay-trace: 40.26 / 40.44 / 40.97

noise-delay-full: 40.36 / 40.62 / 40.67

impulse-channels-trace: 40.47 / 40.53 / 40.56

impulse-channels-full: 40.45 / 40.58 / 40.73

impulse-delay-trace: 40.47 / 40.48 / 40.52

impulse-delay-full: 40.60 / 40.60 / 40.64